Multivariate Time Series Modeling for Comparative Forecasting of Oil Prices

A Case Study

1. Introduction & Background

Crude oil is a primary commodity in the global economy, influencing inflation, trade balances, and policy making. Accurate prediction of oil prices remains elusive due to nonlinear interrelations among geopolitical shocks, supply-demand fundamentals, and macroeconomic drivers. Classical univariate models are not adequate, particularly in times of volatility.

This study designs a comprehensive multivariate modeling pipeline with Brent and WTI oil prices as targets. The aim is to benchmark different time series model, which include statistical, machine learning, and deep learning, within one pipeline.

2. Research Aim and Objectives

Aim: To improve oil price forecasting accuracy by designing and comparing multivariate time series models.

Objectives:

1. Develop VARMAX, XGBoost, and LSTM models for crude oil forecasting.
2. Integrate macroeconomic indicators into forecasting.
3. Compare model performance using consistent metrics.

3. Research Questions

- How can multivariate time series models be designed to improve oil price forecasting accuracy?
- Which model performs best under historical volatility and structural shocks?

4. Rational for Selected Models

The three-model selection was made on the basis of covering different learning paradigms:

- **VARMAX** is a standard econometric model well capable of modeling linear interdependencies and policy shocks.
- **XGBoost** is a powerful gradient boosting model especially effective at detecting feature interactions and non-linearities.
- **LSTM** neural networks can model time-varying patterns and fit sequential data such as prices most appropriately.

The selection of VARMAX, XGBoost, and LSTM models is supported both by econometric theory and recent advances in empirical results from forecasting

literature. VARMAX models are widely used in macroeconomics and energy forecasting because they can model interdependencies of multiple time series and policy-enduced exogenous variables (e.g., Kilian & Murphy, 2012). XGBoost, a scalable gradient boosting technique, has had high performance in financial time series tasks through capturing non-linear interactions and variable importance ordering successfully (Chen & Guestrin, 2016). Last but not least, LSTM networks are well suited for sequential data due to their memory cell structure, which has had high value in capturing long-term dependencies in volatile commodity markets (Yu et al., 2008). This hybrid option allows for an even evaluation of statistical power, machine learning adaptability, and deep learning heterogeneity in advanced oil price environments.

5. Research Design

5.1 Research Approach

A quantitative, empirical study structured according to the CRISP-DM framework.

5.2 Data Design

- **Period:** 1960–2024
- **Sources:**
  - World Bank (Oil prices, energy and metal commodities)
  - IMF (Macroeconomic variables)
  - Caldara & Iacoviello GPR Index
- **Variables:**
  - **Dependent:** Brent and WTI Spot Prices
  - **Independent:**
    - **Australian & South African Coal Prices:** Coal is among the most highly demanded fossil fuels for energy production globally. It normally reacts with oil prices since substitution takes place in energy as well as industrial markets. Monitoring coal prices allows for discovery of changes in demand between types of energy (Apergis & Payne, 2014).
    - **US, European, LNG Japanese Natural Gas Prices:** Natural gas freely competes with oil as a fuel for industry, electricity, and heating. Interconnectivity of regional gas markets, especially through LNG, makes gas price movements often a mirror of changes in energy demand and capable of influencing oil prices (Ramberg & Parsons, 2012).

* **Palm Oil:** Palm oil is used in the production of biofuels and is now relevant as a source to replace fossil fuels. Price changes within it can mirror bioenergy demands, which change crude oil consumption patterns (Zhang & Cheng, 2021).
* **Copper:** Frequently described as a "doctor" metal, copper prices are indicative of world industrial demand. Because oil demand is strongly correlated with economic cycles, copper prices are used as an indicator of world economic activity (Yu et al., 2008).
* **Gold:** Historically, a safe-haven commodity, gold prices in aggregate invert with oil during periods of uncertainty or inflation fears. Including gold guarantees attention to investor mood and macro-financial risk (Wang et al., 2013).

5.3 Sampling Design

- **Training:** Jan 1960 – Dec 2015
- **Testing:** Jan 2016 – Dec 2024

6. Descriptive Analysis

The descriptive analysis was a first step in understanding the pattern of the data before model building. Summary statistics gave insights into centering tendencies and variability and guided scaling and normalization choices. Trend analysis identified macroeconomic-driven long-term cycles of prices, and STL decomposition confirmed seasonality and cyclicality—requiring time-aware models. Correlation matrices enabled interdependencies between variables and guided feature selection for multivariate modeling. Rolling volatility plots revealed grouping tendencies characteristic of oil markets, alerting us to instability periods to be handled by the models. Finally, imputation of missing values and outliers ensured data completeness and purity and reduced the threat of model training on unbalanced information and enhanced forecast accuracy.

6.1 Summary Statistics

Summary statistics such as mean, standard deviation, skewness, and kurtosis provide a preliminary view of the data distribution. This allows for screening for problems like high variance, non-normality, or outliers, which guides preprocessing options such as scaling, transformation, or winsorization.

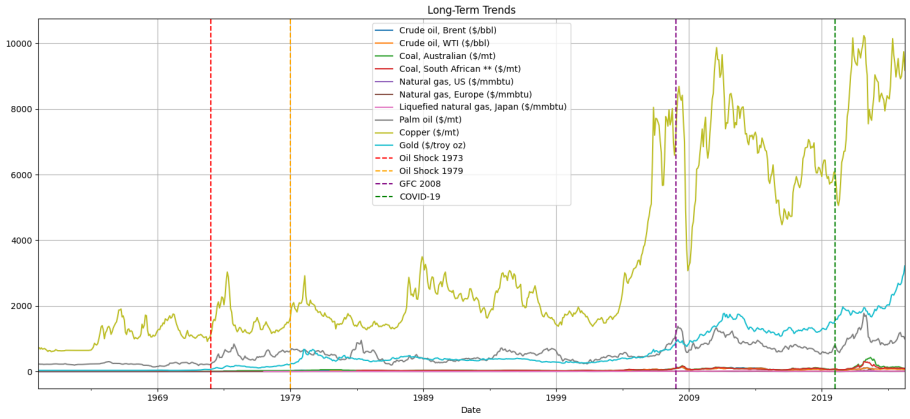
6.2 Trend Analysis

Analysis of long-term trends in Brent and WTI highlights structural change, e.g., owing to geopolitics uncertainty, economic crises, or revolutions in tech-

Table 1: Descriptive Summary Statistics of Variables

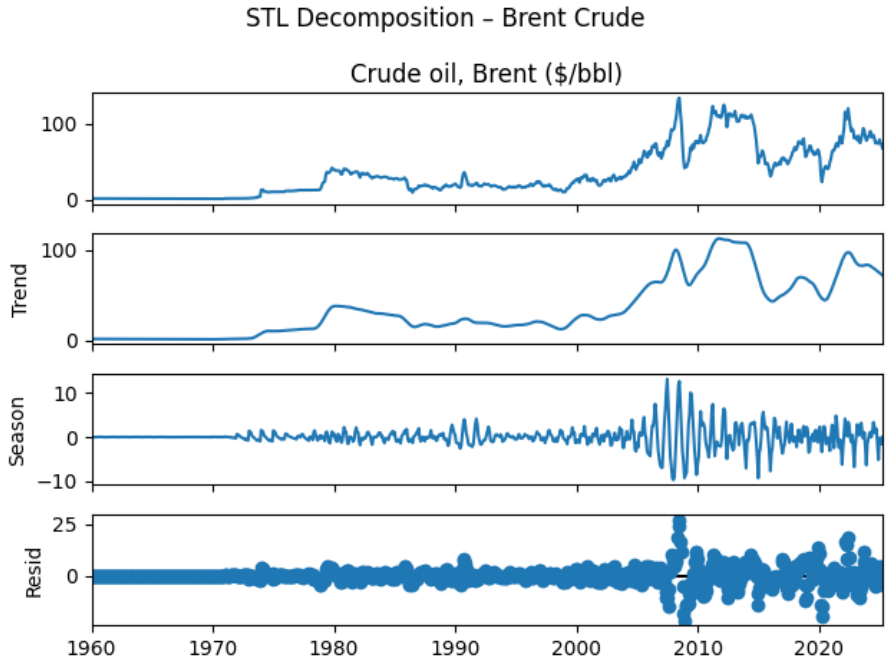
...1	count	mean	std	min	25%	50%	75%	max	skewness	kurtosis
Crude oil, Brent ($/bbl)	784	34.43	32.56	1.210	11.51	22.95	54.63	133.9	1.050	0.129
Crude oil, WTI ($/bbl)	520	46.16	28.57	11.310	20.71	33.26	69.64	133.9	0.744	-0.590
Coal, Australian ($/mt)	664	58.04	54.98	7.800	29.25	38.89	69.43	430.8	3.467	16.362
Coal, South African ** ($/mt)	496	60.79	43.56	21.250	30.49	45.59	83.65	302.0	2.347	8.099
Natural gas, US ($/mmbtu)	784	2.41	2.17	0.140	0.55	2.10	3.03	13.5	1.569	3.512
Natural gas, Europe ($/mmbtu)	784	4.83	6.21	0.384	1.74	3.07	5.92	70.0	4.654	32.739
Liquefied natural gas, Japan ($/mmbtu)	580	7.23	4.43	2.720	3.64	5.55	10.05	23.7	1.118	0.289
Palm oil ($/mt)	784	539.55	282.78	141.730	307.50	494.00	688.00	1777.0	1.014	1.048
Copper ($/mt)	784	3328.26	2692.39	606.710	1407.94	1971.80	5671.36	10230.9	1.046	-0.359
Gold ($/troy oz)	784	593.10	608.89	34.940	151.34	374.64	922.79	3217.6	1.375	1.237

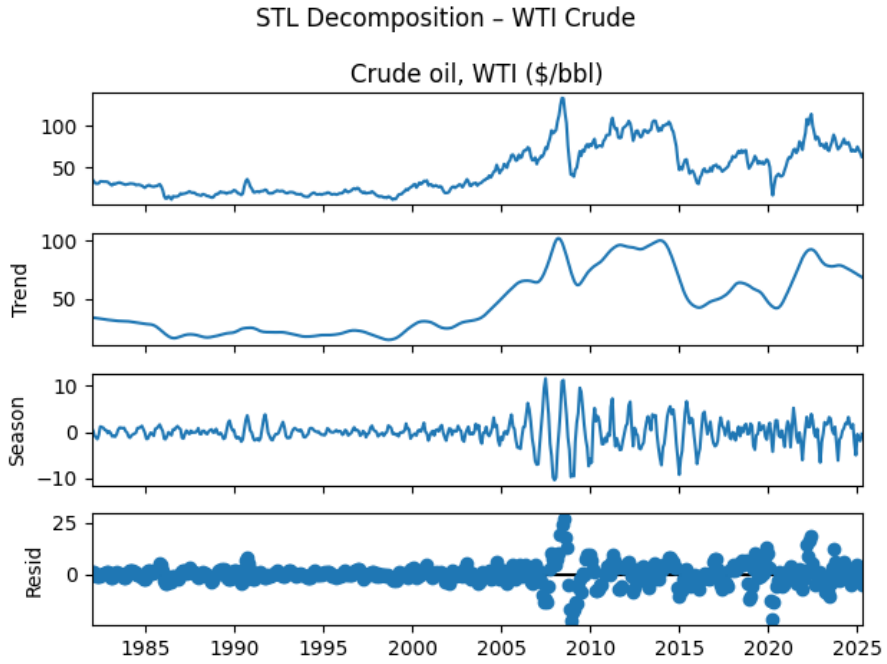
nology. Such evidence supports the need for regime-variant models and price non-stationarity.



6.3 Seasonality & Cyclicality

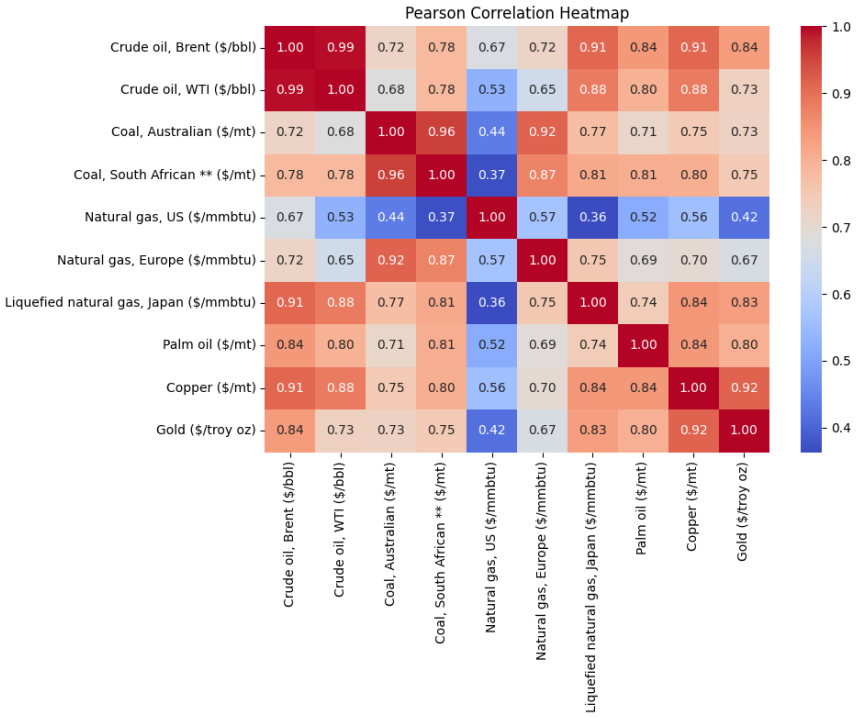
STL decomposition is used to break down the series into trend, seasonal, and residuals. This is done to determine whether the crude oil price has normal cycles or seasonality factors, which are critical in determining models that reflect such temporal patterns, like LSTM or SARIMA.

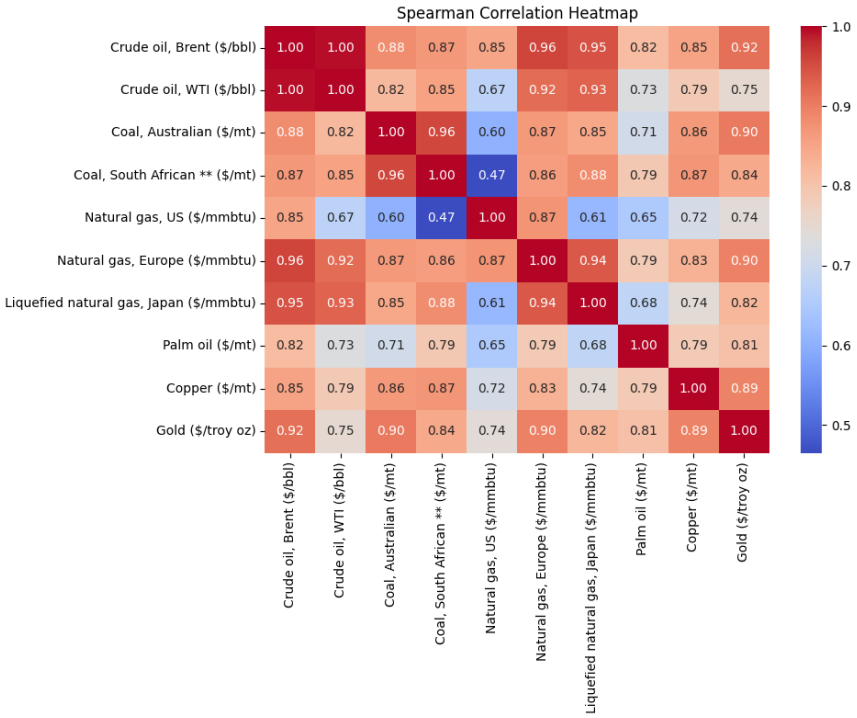

STL Decomposition – Brent Crude
Crude oil, Brent ($/bbl)
Trend
Season
Resid
1960
1970
1980
1990
2000
2010
2020


STL Decomposition – WTI Crude
Crude oil, WTI ($/bbl)
Trend
Season
Resid
1985
1990
1995
2000
2005
2010
2015
2020
2025

6.4 Correlation Analysis

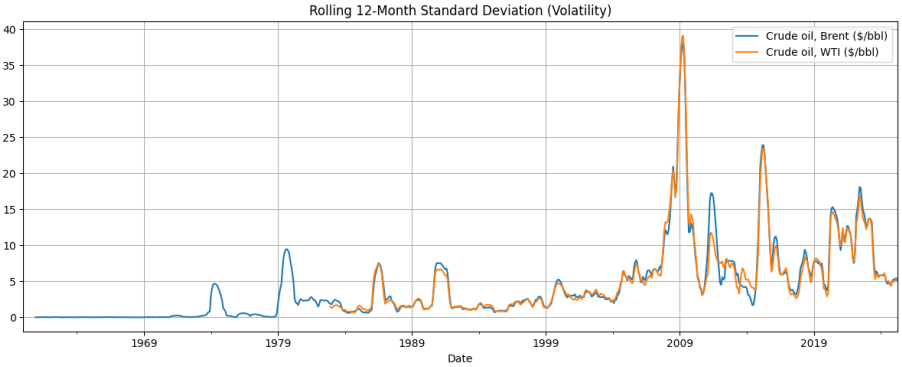
Correlation heatmaps (Pearson and Spearman) reveal correlations among crude oil and any other macroeconomic variable or commodity. High correlations signal possible multicollinearity or predictive relationships, which determine variable choice and model design, particularly when applying multivariate models like VARMAX.





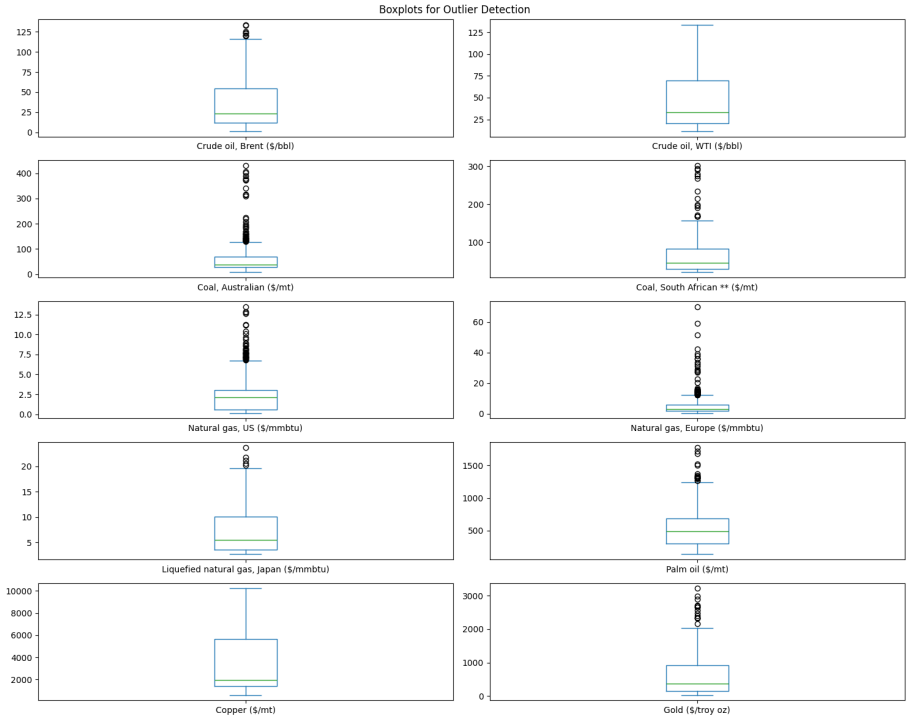
6.5 Volatility Analysis

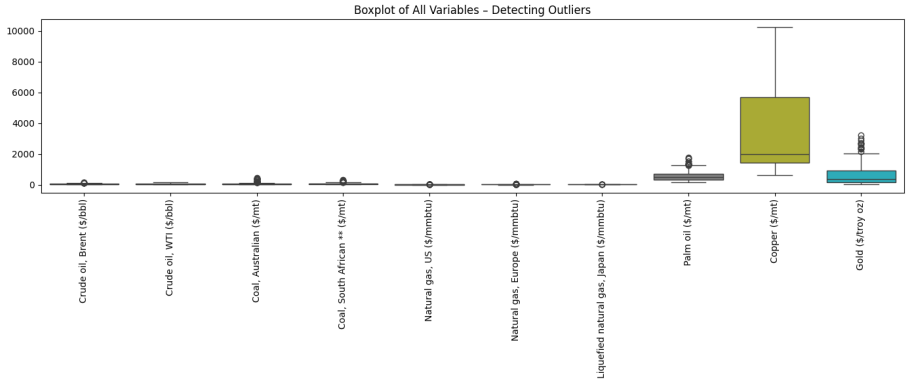
Rolling standard deviation plots help to discover periods of volatility clustering—a feature of energy markets. Distinguishing when and how volatility shifts over time guarantees that the models are insensitive to heteroskedastic behavior, threatening forecasting accuracy.



6.6 Missing & Outlier Detection

Missing values and outliers can bias model training and lead to incorrect predictions. The current step prevents failure of forward/backward fill operations and winsorization. The visual tests ensure that the integrity of the data is preserved after imputation, so that all models learn from homogeneous inputs.





7. Methodology

7.1 Data Preparation

- Missing data: forward and backward fill
- Winsorization: trim 1% tails

Table 2: ADF Test Results for Stationarity

Variable	ADF p-value
Crude oil, Brent ($/bbl)	0.392
Crude oil, WTI ($/bbl)	0.348
Coal, Australian ($/mt)	0.080
Coal, South African ** ($/mt)	0.162
Natural gas, US ($/mmbtu)	0.087
Natural gas, Europe ($/mmbtu)	0.246
Liquefied natural gas, Japan ($/mmbtu)	0.493
Palm oil ($/mt)	0.187
Copper ($/mt)	0.741
Gold ($/troy oz)	0.997

- Normality test: Shapiro

Table 3: Shapiro-Wilk Test Results for Normality

Variable	**Shapiro p-value**
Crude oil, Brent ($/bbl)	7.05×10^{-20}
Crude oil, WTI ($/bbl)	5.50×10^{-19}
Coal, Australian ($/mt)	9.64×10^{-31}
Coal, South African ($/mt)	5.24×10^{-27}
Natural gas, US ($/mmbtu)	1.00×10^{-23}
Natural gas, Europe ($/mmbtu)	4.85×10^{-31}
LNG, Japan ($/mmbtu)	4.20×10^{-21}
Palm oil ($/mt)	1.27×10^{-13}
Copper ($/mt)	6.84×10^{-21}
Gold ($/troy oz)	5.30×10^{-24}

- Autocorrelation: Durbin-Watson

Table 4: Durbin-Watson Test for Autocorrelation

Variable	**Durbin-Watson Stat**
Crude oil, Brent ($/bbl)	0.006
Crude oil, WTI ($/bbl)	0.006
Coal, Australian ($/mt)	0.014
Coal, South African ($/mt)	0.013
Natural gas, US ($/mmbtu)	0.021
Natural gas, Europe ($/mmbtu)	0.024
LNG, Japan ($/mmbtu)	0.003
Palm oil ($/mt)	0.005
Copper ($/mt)	0.003
Gold ($/troy oz)	0.001

- Lag selection: AIC from VAR model optimal lag = 12
- Scaling: Min-Max scaling

7.2 Model Development

1. VARMAX

- Endogenous: Brent, WTI
- Exogenous: macroeconomic indicators

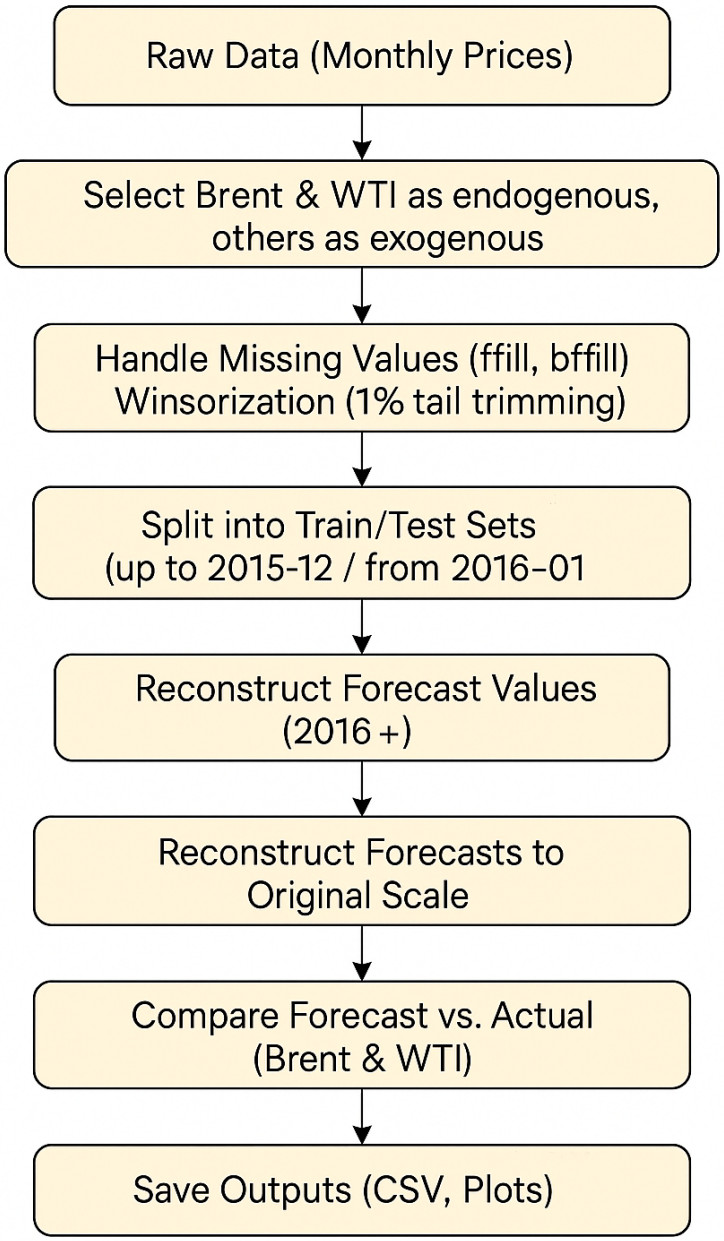
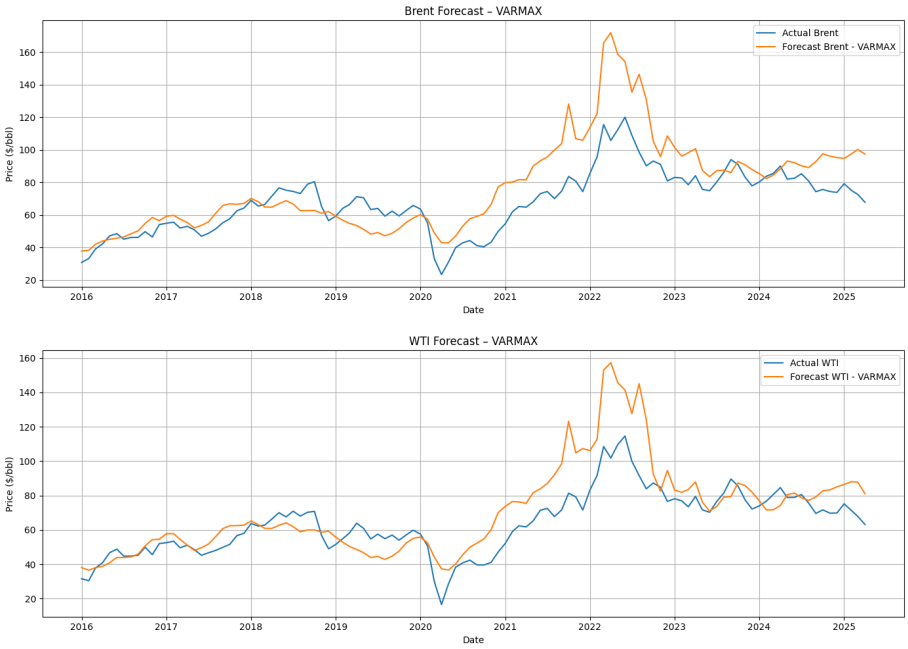
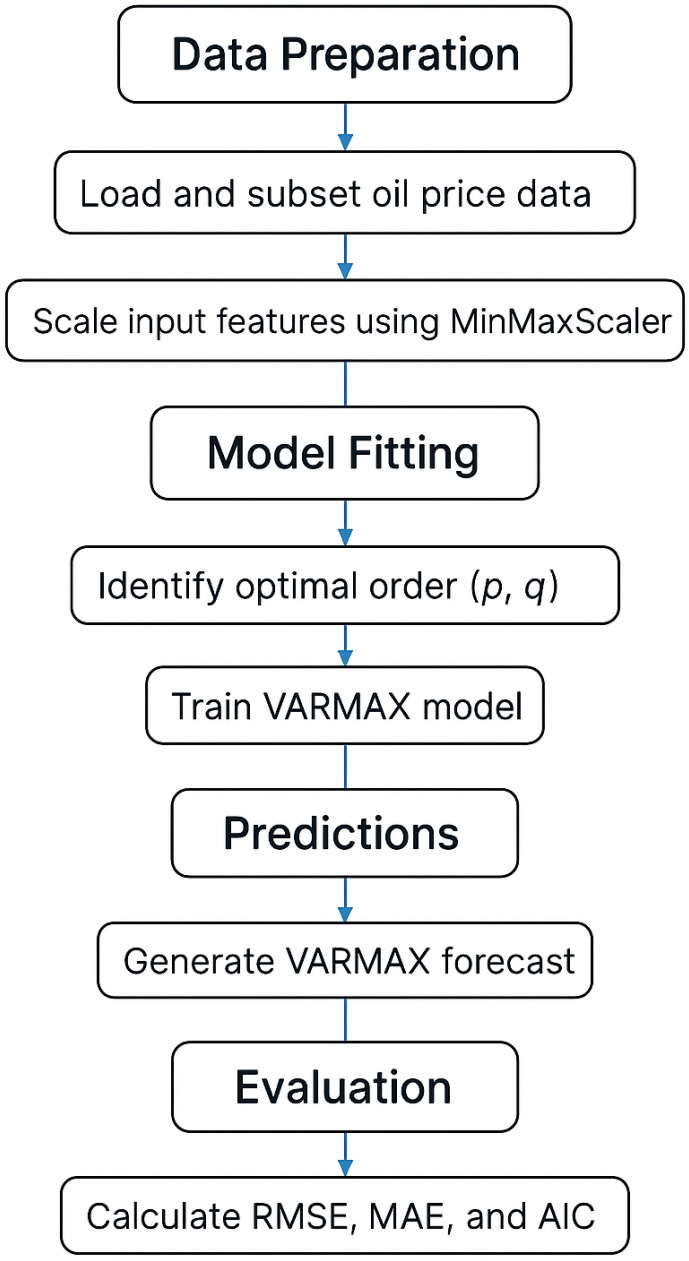

Raw Data (Monthly Prices)
Select Brent & WTI as endogenous, others as exogenous
Handle Missing Values (ffill, bffill) Winsorization (1% tail trimming)
Split into Train/Test Sets (up to 2015-12 / from 2016-01
Reconstruct Forecast Values (2016 +)
Reconstruct Forecasts to Original Scale
Compare Forecast vs. Actual (Brent & WTI)
Save Outputs (CSV, Plots)

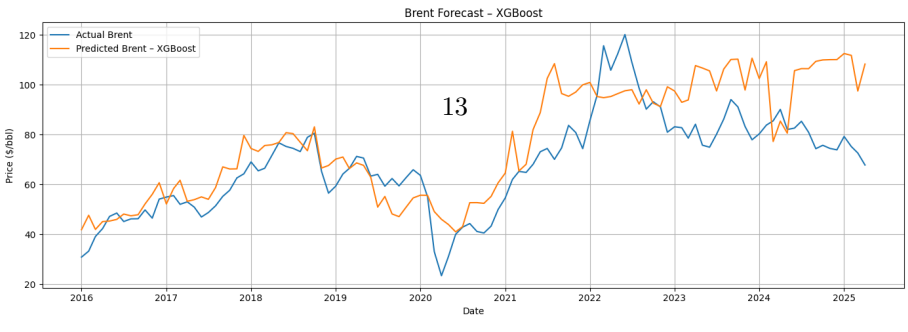
Table 5: Forecasted Values using VARMAX (First 10 Rows)

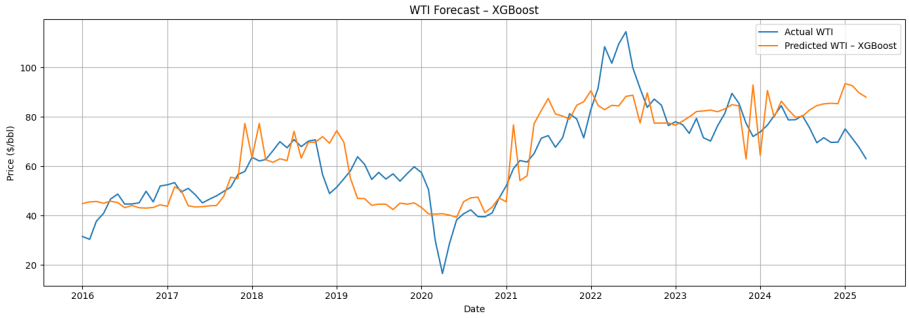
Date	Crude oil, Brent ($/bbl)	Crude oil, WTI ($/bbl)
2016-01-01	37.8	38.0
2016-02-01	38.3	36.5
2016-03-01	41.9	38.0
2016-04-01	44.0	38.8
2016-05-01	45.0	40.8
2016-06-01	45.7	44.0
2016-07-01	46.4	43.9
2016-08-01	48.4	44.3
2016-09-01	50.2	45.9
2016-10-01	54.8	50.7

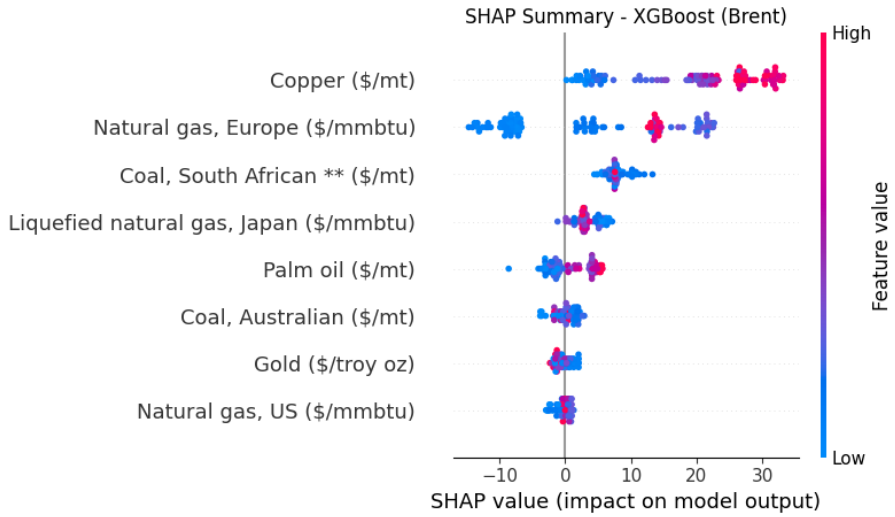


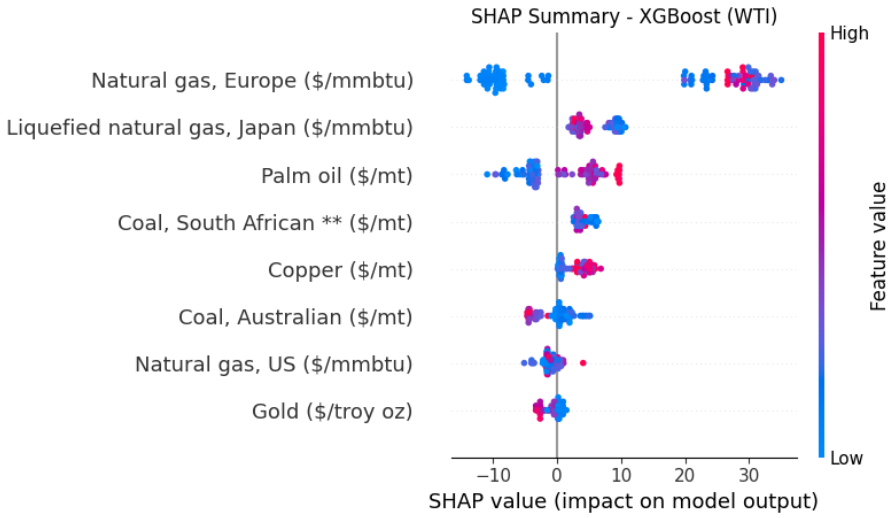
2. XGBoost


Data Preparation
Load and subset oil price data
Scale input features using MinMaxScaler
Model Fitting
Identify optimal order (p, q)
Train VARMAX model
Predictions
Generate VARMAX forecast
Evaluation
Calculate RMSE, MAE, and AIC


Brent Forecast – XGBoost
Actual Brent
Predicted Brent – XGBoost
Price ($/bbl)
Date
120
100
80
60
40
20
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025


WTI Forecast – XGBoost
Actual WTI
Predicted WTI – XGBoost
Price ($/bbl)
Date
100
80
60
40
20
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025


SHAP Summary - XGBoost (Brent)
Copper ($/mt)
Natural gas, Europe ($/mmbtu)
Coal, South African ** ($/mt)
Liquefied natural gas, Japan ($/mmbtu)
Palm oil ($/mt)
Coal, Australian ($/mt)
Gold ($/troy oz)
Natural gas, US ($/mmbtu)
−10
0
10
20
30
SHAP value (impact on model output)
High
Feature value
Low


SHAP Summary - XGBoost (WTI)
Natural gas, Europe ($/mmbtu)
Liquefied natural gas, Japan ($/mmbtu)
Palm oil ($/mt)
Coal, South African ** ($/mt)
Copper ($/mt)
Coal, Australian ($/mt)
Natural gas, US ($/mmbtu)
Gold ($/troy oz)
−10
0
10
20
30
SHAP value (impact on model output)
High
Feature value
Low

3. LSTM

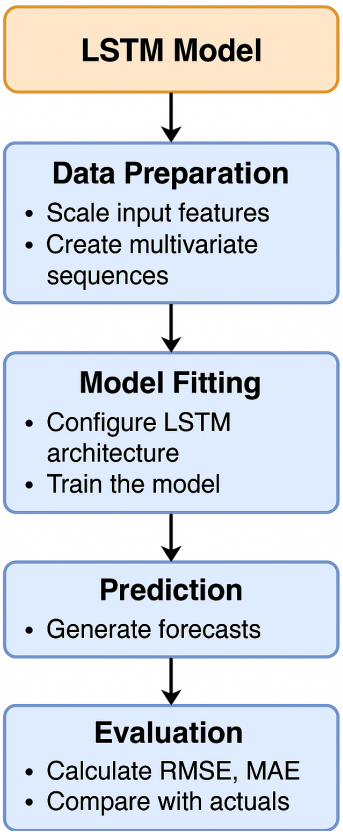
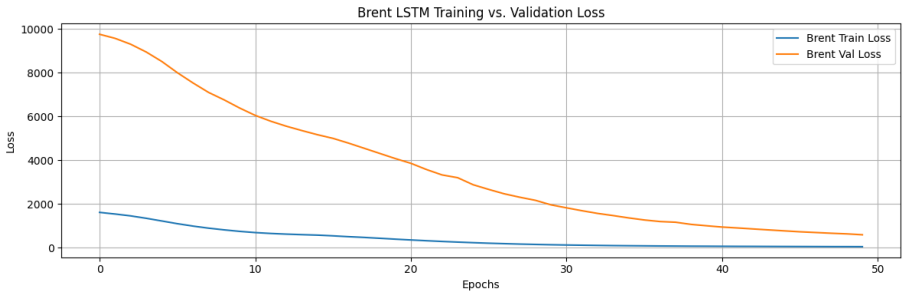
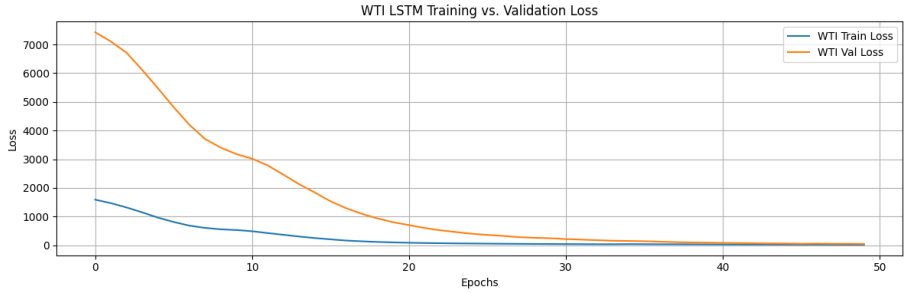
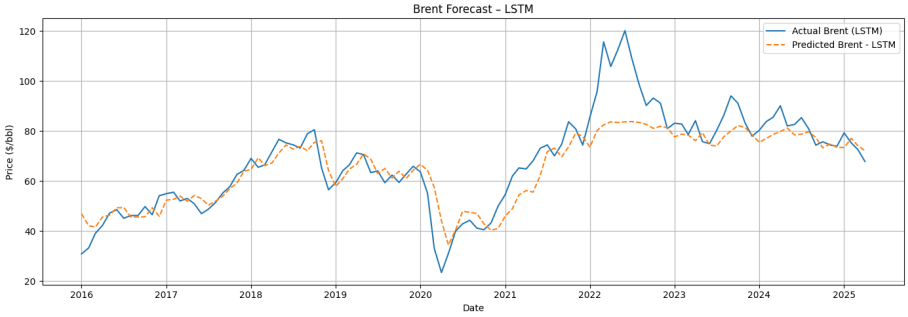
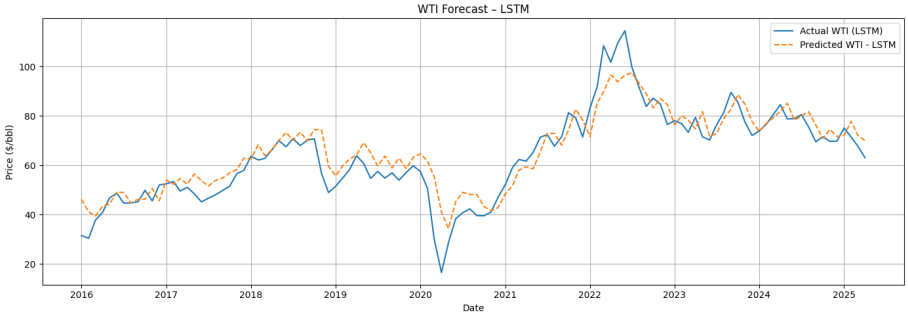


Figure 1: LSTM Model Flowchart



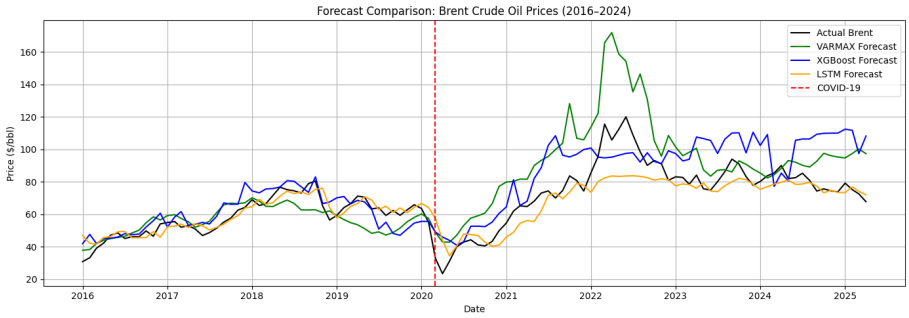

WTI LSTM Training vs. Validation Loss
WTI Train Loss
WTI Val Loss
Loss
Epochs


Brent Forecast – LSTM
Actual Brent (LSTM)
Predicted Brent - LSTM
Price ($/bbl)
Date


WTI Forecast – LSTM
Actual WTI (LSTM)
Predicted WTI - LSTM
Price ($/bbl)
Date

8. Forecast Comparison

Brent Forecasts:



WTI Forecasts:

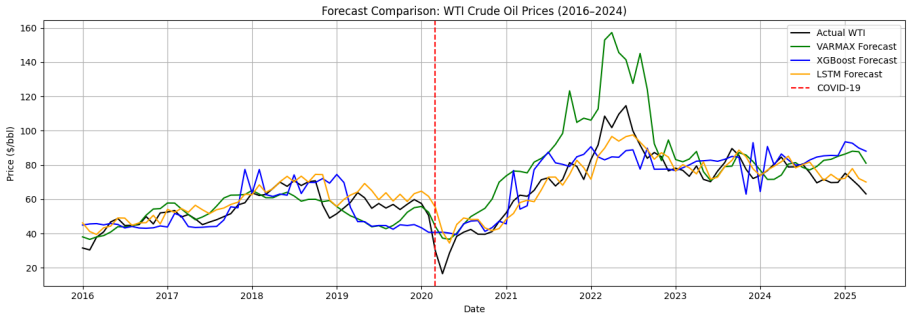


Figure 2: Actual vs. Forecast:WTI

9. Evaluation & Diagnostics

9.1 Forecast Accuracy

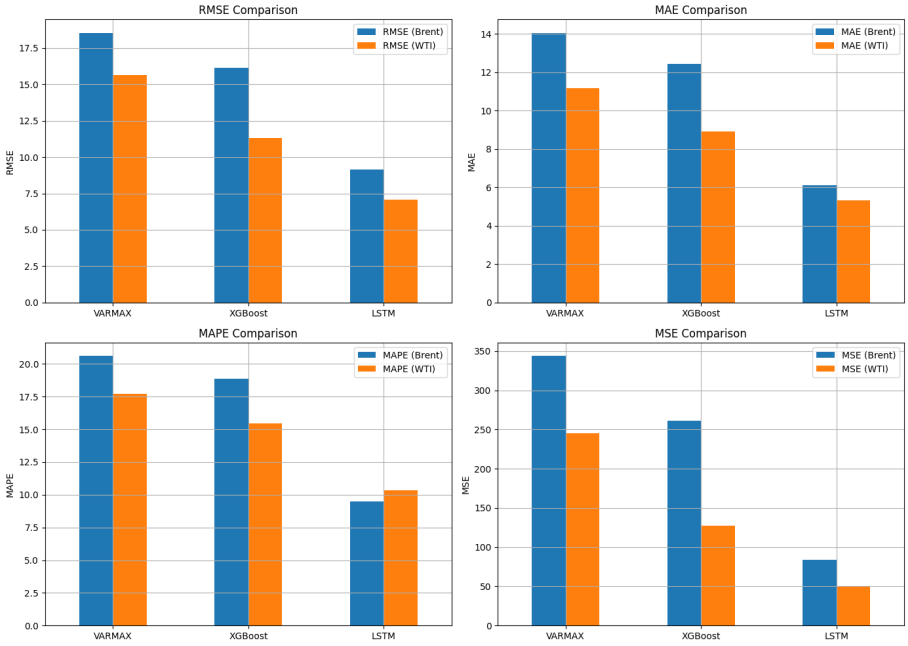


Table 6: Evaluation Metrics - Brent

Model	RMSE	MSE	MAE	MAPE
VARMAX	18.54	343.6	14.04	20.60
XGBoost	16.15	260.9	12.45	18.85
LSTM	9.14	83.6	6.12	9.49

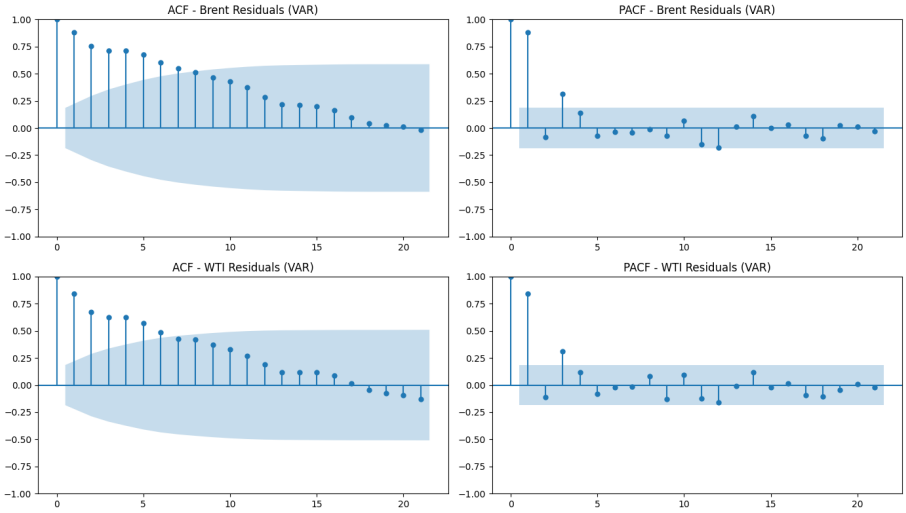
Table 7: Evaluation Metrics - WTI

Model	RMSE	MSE	MAE	MAPE
VARMAX	15.67	245	11.17	17.7
XGBoost	11.29	128	8.92	15.5
LSTM	7.07	50	5.31	10.3

Table 8: Breusch-Pagan Test for Heteroskedasticity

Commodity	LM stat	LM p-value	F stat	F p-value
Brent	0.897	0.639	0.44	0.645
WTI	5.148	0.076	2.63	0.077

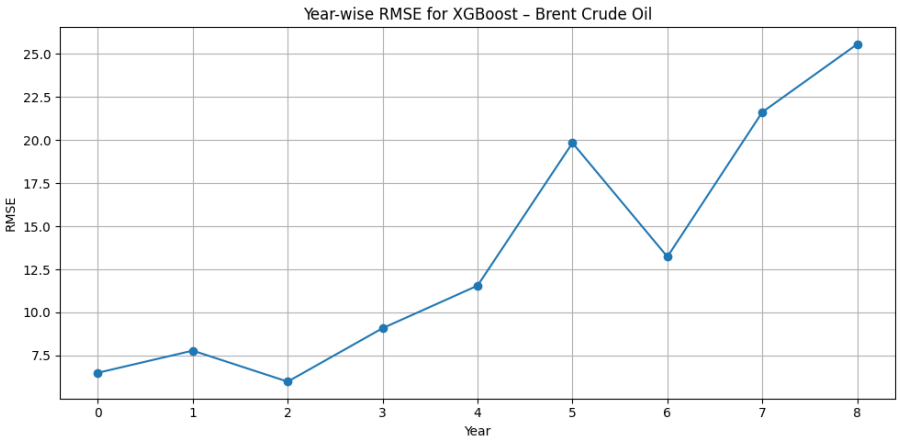
9.2 Residual Diagnostics



9.3 Year-Wise Stability

Table 9: Year-wise RMSE for XGBoost (Brent)

Year	Brent RMSE
2016	6.50
2017	7.79
2018	5.99
2019	9.09
2020	11.55
2021	19.83
2022	13.23
2023	21.62
2024	25.56



10. Outcomes

- **Best overall performance: LSTM**, with the lowest RMSE and MAE for Brent and WTI forecasts, outperforming other models in accuracy and consistency.
- **Average performer: XGBoost**, with error rates not too excessive and was doing well without the requirements of sequence modeling.
- **Most error-unstable model: VARMAX**, although statistically founded, had the highest rates of errors and could have struggled with nonlinear and error-increasing patterns.

LSTM does the best among all others with a Brent RMSE of 9.14 and WTI RMSE of 7.07, demonstrating how well it can capture temporal relationships and allow nonlinearities. Its significantly lower MAE and MAPE also reflect how it makes consistently accurate predictions, particularly useful during times of volatility and regime change.

XGBoost generates similar performance, particularly for WTI, with RMSE of 11.29, leveraging its multicollinearity and nonlinear interaction strength. Although it is not as accurate as LSTM, it is a reliable and cost-effective choice.

VARMAX, though grounded in economic theory, performs worst according to error metrics. Its higher RMSE (18.54 for Brent and 15.67 for WTI) and MAE suggest that it fails to excel in capturing complex modern oil price dynamics, especially when exogenous shocks are nonlinear or abrupt.

Interestingly, the Breusch–Pagan test shows no significant evidence of heteroskedasticity in Brent ($p = 0.639$), whereas marginal heteroskedasticity

in WTI ($p = 0.077$). This is partially the reason why simpler models like VARMAX underperform in turbulent conditions.

12. References

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